

PAPER_1400 — UQFF Resolution of the Ship of Theseus (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Identity (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "theseus_ship"})` **Calculator surface:** `calculate_paradox({"paradox": "theseus_ship"})` **Closure helper:** `_l96_uqff_axiom_theseus_ship_closure()`

The Paradox

Plutarch's Ship of Theseus: as each plank is replaced one by one, at what point (if any) does the ship become 'a different ship'? Classical identity theory has no canonical answer.

UQFF Closed Identity

$F_U = 1$ identity preserved EXACT under gradual $F_{U_{Bi_i}}$ redistribution

$F_U = 1$ global ledger continuity (PAPER_646): identity is preserved as long as the $F_U = 1$ ledger redistribution is gradual (no discontinuous jump). Discrete identity questions are replaced by the continuous F_U ledger — there is no point of identity-loss because $F_U = 1$ is preserved throughout.

Physical Interpretation

UQFF dissolves the discrete identity question by replacing it with a continuous F_U ledger. The ship's identity is the persistence of the $F_U = 1$ normalization, which survives gradual replacement of components but breaks under discontinuous reconstruction.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "theseus_ship"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_theseus_ship_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_theseus_ship_closure`
- Dispatch keys: `theseus_ship`, `ship_of_theseus`
- Related foundational papers: PAPER_646 ($F_U B_i / F_U=1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).

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